cl_nav2z::CbNavigateForward

+ CbNavigateForward()
+ CbNavigateForward()
+ CbNavigateForward()
+ ~CbNavigateForward()
+ onEntry()
+ onExit()
+ setForwardDistance()
+ CbNavigateForward()
+ CbNavigateForward()
+ CbNavigateForward()
+ ~CbNavigateForward()
+ onEntry()
+ onExit()
+ setForwardDistance()

cl_nav2z::CbNav2ZClientBehaviorBase

+ ~CbNav2ZClientBehaviorBase()
+ onStateOrthogonalAllocation()
+ onOrthogonalAllocation()
+ ~CbNav2ZClientBehaviorBase()
+ onOrthogonalAllocation()
+ onStateOrthogonalAllocation()
# sendGoal()
# cancelGoal()
# onNavigationSucceeded()
# onNavigationAborted()
# onNavigationCancelled()
# onNavigationResult()
# onNavigationActionSuccess()
# onNavigationActionAbort()
# sendGoal()
# cancelGoal()
# isOwnActionResponse()
# onNavigationResult()
# onNavigationActionSuccess()
# onNavigationActionAbort()

smacc2::SmaccAsyncClient
Behavior

+ onOrthogonalAllocation()
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()

smacc2::ISmaccClientBehavior

+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onOrthogonalAllocation()
- onStateOrthogonalAllocation()

smacc2::ISmaccStateMachine

+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExitting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration
Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()

-stateMachine_

smacc2::SignalDetector

+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()

-smaccStateMachine_

smacc2::ISmaccOrthogonal

+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()

-stateMachine_

smacc2::ISmaccComponent

+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onOrthogonalAllocation()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()

#stateMachine_

smacc2::ISmaccClient

+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onOrthogonalAllocation()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()

-stateMachine_

SharedPtr

#nh_

smacc2::ISmaccState

+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition
TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()

#node_

SharedPtr

#timer_

cl_nav2z::odom_tracker
::CpOdomTracker

+ CpOdomTracker()
+ processNewPose()
+ odomMessageCallback()
+ update()
+ setWorkingMode()
+ setPublishMessages()
+ pushPath()
+ pushPath()
+ popPath()
+ clearPath()
+ setStartPoint()
+ setStartPoint()
+ setCurrentMotionGoal()
+ setCurrentPathName()
+ getCurrentMotionGoal()
+ getPath()
+ logStateString()
+ updateParameters()
+ setOdomFrame()
+ CpOdomTracker()
+ processNewPose()
+ odomMessageCallback()
+ update()
+ setWorkingMode()
+ setPublishMessages()
+ pushPath()
+ pushPath()
+ popPath()
+ clearPath()
+ setStartPoint()
+ setStartPoint()
+ setCurrentMotionGoal()
+ setCurrentPathName()
+ getCurrentMotionGoal()
+ getPath()
+ logStateString()
+ updateParameters()
+ setOdomFrame()
# onInitialize()
# updateConfiguration()
# rtPublishPaths()
# updateRecordPath()
# updateClearPath()
# updateAggregatedStackPath()
# parametersCallback()
# onInitialize()
# updateConfiguration()
# rtPublishPaths()
# updateRecordPath()
# updateClearPath()
# updateAggregatedStackPath()
# parametersCallback()

#robotPoseTimer_

SharedPtr

#stateMachinePub_

SharedPtr

#stateMachineStatusPub_

-statusPub_

SharedPtr

#transitionLogPub_

SharedPtr

#eventsLogPub_

SharedPtr

#transitionHistoryService_

vector< smacc2::ISmaccState * >

#currentState_

shared_ptr< smacc2
::introspection::SmaccState
Info >

#currentStateInfo_

smacc2::introspection
::SmaccStateInfo

+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()

+parentState_

SmaccStatus

#status_msg_

map< std::string, std
::shared_ptr< smacc2
::ISmaccOrthogonal > >

#orthogonals_

vector< boost::signals2
::scoped_connection >

#longLivedSignalConnections_

shared_ptr< smacc2
::introspection::SmaccState
MachineInfo >

#stateMachineInfo_

+stateMachine_

recursive_mutex

-eventQueueMutex_
-m_mutex_

StateMachineInternalAction

-stateMachineCurrentAction

map< void *, std::shared
_ptr< smacc2::CallbackCounterSemaphore > >

-stateCallbackConnections

map< std::string, std
::pair< std::function
< std::string()>, boost
::any > >

-globalData_

vector< smacc2_msgs
::msg::SmaccTransitionLogEntry >

-transitionLogHistory_

SMRunMode

-runMode_

-signalDetector_

vector< smacc2::ISmaccUpdatable * >

-updatableClients_

vector< std::vector
< smacc2::ISmaccUpdatable * > >

-updatableStateElements_

atomic< int64_t >

-lastState_

double

-loop_rate_hz

#clearAngularDistanceThreshold
_

#clearPointDistanceThreshold

_

#recordAngularDistanceThreshold
_

#recordPointDistanceThreshold_

atomic< bool >

-end_
-initialized_

SmaccFifoScheduler *

-scheduler_

processor_handle

-processorHandle_

thread

-signalDetectorThread_

ExecutionModel

-executionModel_

int64_t

-stateSeqCounter_

-currentState

#parentState_

shared_ptr< rclcpp
::Logger >

#logger_

vector< std::shared
_ptr< smacc2::StateReactor > >

#stateReactors_

vector< std::shared
_ptr< smacc2::SmaccEventGenerator > >

#eventGenerators_ #stateInfo_

std::enable_shared
_from_this< SmaccStateInfo >

map< const std::type
_info *, std::vector
< smacc2::introspection
::ClientBehaviorInfoEntry > >

+staticBehaviorInfo

map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccStateReactor
Info > > >

+stateReactorsInfo

map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccEventGenerator
Info > > >

+eventGeneratorsInfo

int

+depth_
+stateIndex_

string

+demangledStateName
+fullStateName

cl_nav2z::ClNav2Z

+ ClNav2Z()
+ ~ClNav2Z()
+ onComponentInitialization()
+ ClNav2Z()
+ ~ClNav2Z()

-actionServerName_

smacc2::client_bases
::SmaccActionClientBase
< nav2_msgs::action::NavigateTo
Pose >

+ SmaccActionClientBase()
+ SmaccActionClientBase()
+ ~SmaccActionClientBase()
+ getClientBase()
+ onInitialize()
+ waitForServer()
+ postResultEvent()
+ onOrthogonalAllocation()
+ onSucceeded()
+ onSucceeded()
+ onAborted()
+ onAborted()
+ onCancelled()
+ onCancelled()
+ cancelGoal()
+ sendGoal()
+ getEventLabel()
# onFeedback()
# onResult()

+action_endpoint_

smacc2::client_bases
::ISmaccActionClient

+ ISmaccActionClient()
+ ~ISmaccActionClient()
+ getNamespace()
+ cancelGoal()
+ getClientBase()
+ getName()

#name_

smacc2::client_bases
::SmaccActionClientBase
< ActionType >

+ SmaccActionClientBase()
+ SmaccActionClientBase()
+ ~SmaccActionClientBase()
+ getClientBase()
+ onInitialize()
+ waitForServer()
+ postResultEvent()
+ onOrthogonalAllocation()
+ onSucceeded()
+ onSucceeded()
+ onAborted()
+ onAborted()
+ onCancelled()
+ onCancelled()
+ cancelGoal()
+ sendGoal()
+ getEventLabel()
# onFeedback()
# onResult()

+action_endpoint_

#currentPathName_
#odomFrame_
#odomTopicName_

cl_nav2z::CpPose

+ CpPose()
+ CpPose()
+ onInitialize()
+ update()
+ waitTransformUpdate()
+ toPoseMsg()
+ toPoseStampedMsg()
+ getYaw()
+ getX()
+ getY()
+ getZ()
+ setReferenceFrame()
+ getReferenceFrame()
+ getFrameId()
+ freezeReferenceFrame()
+ unfreezeReferenceFrame()

-poseFrameName_
-referenceFrame_

cl_nav2z::Pose

+ Pose()
+ Pose()
+ onInitialize()
+ update()
+ waitTransformUpdate()
+ toPoseMsg()
+ toPoseStampedMsg()
+ getYaw()
+ getX()
+ getY()
+ getZ()
+ setReferenceFrame()
+ getReferenceFrame()
+ getFrameId()
+ freezeReferenceFrame()
+ unfreezeReferenceFrame()

-poseFrameName_
-referenceFrame_

vector< smacc2::introspection
::SmaccTransitionInfo >

+transitions_

vector< std::shared
_ptr< smacc2::introspection
::SmaccStateInfo > >

+children_

type_info *

+tid_

-currentOrthogonal

-orthogonal_

vector< std::shared
_ptr< smacc2::ISmaccClient > >

#clients_

vector< std::vector
< std::shared_ptr< smacc2
::ISmaccClientBehavior > > >

-clientBehaviors_

mutex

-mutex_

smacc2::client_core
_components::CpActionClient
< nav2_msgs::action::NavigateToPose >

+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()

-actionMutex_

smacc2::client_core
_components::CpActionClient
< ActionType >

+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()

-actionMutex_

#m_mutex_

-m_mutex_ -m_mutex_

optional< std::future
< int > >

-onEntryThread_
-onExitThread_

function< void()>

-postFailureEventFn
_

-postFinishEventFn
_

-postSuccessEventFn_

smacc2::SmaccSignal
< void()>

-onFailure_
-onFinished_
-onSuccess_

+onGoalAccepted_
+onGoalRejected_

+onGoalAccepted_
+onGoalRejected_

boost::signals2::signal
< void(), optional_last
_value< typename boost::
function_traits< void()>
::result_type >, int, std
::less< int >, function< void
()>, typename extended_signature
< function_traits< void()>::arity,
void()>::function_type, boost::signals2::mutex >

smacc2::SmaccSignal
< Signature, Combiner,
Group, GroupCompare,
SlotFunction, ExtendedSlotFunction,
Mutex >

< void()>

smacc2::SmaccSignal
< void(const WrappedResult &)>

< void(const WrappedResult &)>

smacc2::SmaccSignal
< void(const Feedback &)>

< void(const Feedback &)>

boost::signals2::signal
< Signature, optional
_last_value< typename boost
::function_traits< Signature
>::result_type >, int, std
::less< int >, function< Signature
>, typename extended_signature<
function_traits< Signature >::arity,
Signature >::function_type, boost::
signals2::mutex >

bool

-isShutdownRequested_

#publishMessages

+isInitialized +isInitialized

cl_nav2z::components
::CpNav2ActionInterface

+ CpNav2ActionInterface()
+ ~CpNav2ActionInterface()
+ sendNavigationGoal()
+ sendGoal()
+ cancelNavigation()
+ isNavigationServerReady()
+ waitForNavigationServer()
+ onComponentInitialization()
+ onInitialize()
+ getActionClient()
+ onNavigationSucceeded()
+ onNavigationAborted()
+ onNavigationCancelled()
+ onNavigationFeedback()
- onNavigationSuccessCallback()
- onNavigationAbortedCallback()
- onNavigationCancelledCallback()
- onNavigationFeedbackCallback()

#nav2ActionInterface_

#owner_

map< smacc2::ComponentKey,
std::shared_ptr< smacc2
::ISmaccComponent > >

#components_

+onNavigationAborted
_

+onNavigationCancelled
_

+onNavigationSucceeded_

+onActionAborted_
+onActionCancelled

_

+onActionSucceeded_

+onActionAborted_
+onActionCancelled

_

+onActionSucceeded_

+onAborted_
+onCancelled_
+onSucceeded_

+onAborted_
+onCancelled_
+onSucceeded_

boost::signals2::signal
< void(const WrappedResult
&), optional_last_value
<typename boost::function
_traits<void(const WrappedResult
&)>::result_type>, int, std::less
<int>, function<void(const WrappedResult
&)>, typename extended_signature<function
_traits<void(const WrappedResult &)>::arity,
void(const WrappedResult &)>::function_type,
boost::signals2::mutex >

+onNavigationFeedback_

+onActionFeedback_

+onActionFeedback_

boost::signals2::signal
< void(const Feedback
&), optional_last_value
< typename boost::function
_traits< void(const Feedback
&)>::result_type >, int, std
::less< int >, function< void
(const Feedback &)>, typename extended
_signature< function_traits< void(const
Feedback &)>::arity, void(const Feedback
&)>::function_type, boost::signals2::mutex >

function< void(const
WrappedResult &)>

+postNavigationAbortedEvent
+postNavigationCancelledEvent
+postNavigationSuccessEvent

+postAbortedEvent
+postCancelledEvent
+postSuccessEvent

+postAbortedEvent
+postCancelledEvent
+postSuccessEvent

function< void(const
Feedback &)>

+postNavigationFeedbackEvent

+postFeedbackEvent

+postFeedbackEvent

+postFeedbackEvent

+postFeedbackEvent

CpActionClient< ActionType > *

-actionClient_

#actionClient_

optional< std::string >

+actionServerName

+actionServerName

cl_nav2z::CbNavigateForward
Options

+goalChecker_

optional< std::chrono
::milliseconds >

+serverTimeout

+serverTimeout

shared_ptr< ActionClient >

-client_

-client_

optional< std::shared
_future< typename GoalHandle
::SharedPtr > >

-lastRequest_

-lastRequest_

+lastRequest_

+lastRequest_

optional< std::shared
_future< typename rclcpp
_action::Client< nav2_msgs
::action::NavigateToPose >
::CancelResponse::SharedPtr > >

-lastCancelResponse_

FeedbackCallback

-feedbackCallback_

-feedbackCallback_

+feedback_cb

+feedback_cb

< nav2_msgs::action
::NavigateToPose >

optional< std::shared
_future< typename rclcpp
_action::Client< ActionType
>::CancelResponse::SharedPtr > >

-lastCancelResponse_

ResultCode

#nav2zClient_ #navigationResult_

optional< std::shared
_future< typename CancelResponse
::SharedPtr > >

+lastCancelResponse_

+lastCancelResponse_

function< void(WrappedResult)>

+postAbortedEvent
+postCancelledEvent
+postSuccessEvent

+postAbortedEvent
+postCancelledEvent
+postSuccessEvent

SharedPtr

#client_

#client_

< nav2_msgs::action
::NavigateToPose >

SharedPtr

#navigationCallback_

shared_future< std
::shared_ptr< rclcpp
_action::ClientGoalHandle
< nav2_msgs::action::NavigateTo
Pose > > >

#goalHandleFuture_

+options

optional< float >

#forwardDistance_

+forwardSpeed

optional< geometry
_msgs::msg::Quaternion >

+forceInitialOrientation #odomTracker_

SharedPtr

#robotBasePathPub_
#robotBasePathStackedPub_

SharedPtr

#robotPose_ #odomSub_

smacc2::ISmaccUpdatable

+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()

optional< rclcpp::Duration >

-periodDuration_

optional< rclcpp::Time >

+frozenReferenceFrameTime

-lastUpdate_

+frozenReferenceFrameTime

PoseStamped

-pose_ -pose_

shared_ptr< tf2_ros
::Buffer >

-tfBuffer_ -tfBuffer_

shared_ptr< tf2_ros
::TransformListener >

-tfListener_ -tfListener_

mutex

-listenerMutex_ -listenerMutex_

#robotPose_

Path

#aggregatedStackPathMsg
_

#baseTrajectory_

WorkingMode

#workingMode_

vector< nav_msgs::msg
::Path >

#pathStack_

vector< cl_nav2z::odom
_tracker::CpOdomTracker
::PathInfo >

#pathInfos_

OdomTrackerStrategy

#strategy_

optional< geometry
_msgs::msg::PoseStamped >

#goalPose_

#currentMotionGoal_

SharedPtr

#param_callback_handle_